



OPEN Machine learning for early detection and severity classification in people with Parkinson's disease

Juseon Hwang¹, Changhong Youm¹✉, Hwayoung Park², Bohyun Kim², Hyejin Choi¹ & Sang-Myung Cheon³

Early detection of Parkinson's disease (PD) and accurate assessment of disease progression are critical for optimizing treatment and rehabilitation. However, there is no consensus on how to effectively detect early-stage PD and classify motor symptom severity using gait analysis. This study evaluated the accuracy of machine learning models in classifying early and moderate-stages of PD based on spatiotemporal gait features at different walking speeds. A total of 178 participants were recruited, including 103 individuals with PD (61 early-stage, 42 moderate-stage) and 75 healthy controls. Participants performed a walking test on a 24-m walkway at three speeds: preferred walking speed (PWS), 20% faster (HWS), and 20% slower (LWS). Key features—walking speed at PWS, stride length at HWS, and the coefficient of variation (CV) of the stride length at LWS—achieved a classification accuracy of 78.1% using the random forest algorithm. For early PD detection, the stride length at HWS and CV of the stride length at LWS provided an accuracy of 67.3% with Naïve Bayes. Walking at PWS was the most critical feature for distinguishing early from moderate PD, with an accuracy of 69.8%. These findings suggest that assessing gait over consecutive steps under different speed conditions may improve the early detection and severity assessment of individuals with PD.

Keywords Parkinson's disease, Gait, Severity, Motor symptom, Artificial intelligence, Machine learning

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra of the midbrain, which disrupts the function of the basal ganglia circuitry and impairs motor control^{1,2}. One of the key motor symptoms of PD is the reduced ability to perform automated movements, such as walking³. Consequently, individuals with PD often experience significant locomotor difficulties, leading to a loss of independence and reduced quality of life (QoL)⁴. Given these challenges, early management of PD is crucial. Timely intervention may help delay disease progression, alleviate both motor and non-motor symptoms, and enhance the QoL of people with PD, potentially resulting in substantial cost savings for healthcare systems⁵. However, one of the primary obstacles to early intervention is that PD is often diagnosed at an advanced stage⁵. Moreover, distinguishing early-stage PD from normal age-related gait impairments is challenging, as both conditions may present with similar symptoms, such as decreased mobility and impaired balance, due to age-related degeneration in musculoskeletal and neurological functions⁶. The early symptoms of PD are often subtle, leading to delayed diagnosis and treatment until the disease has progressed and symptoms become more pronounced and difficult to manage⁷. Hence, there is growing interest in using gait assessments as a potential diagnostic tool for detecting PD in its early stages. While gait assessments show promise for the early diagnosis of PD^{8–10}, further research is required to refine these methods and improve their accuracy in distinguishing PD from other age-related conditions. Such advancements could enable earlier interventions, improve patient outcomes, and slow disease progression.

Recently, studies have been conducted on the early detection of PD through gait assessment using machine learning (ML) and deep learning algorithms based on artificial intelligence^{11–13}. For instance, Ferreira et al. conducted a gait analysis to classify individuals with early-stage PD (Hoehn and Yahr [H&Y] stages 1 and 2) and healthy controls. The study demonstrated that the support vector machine (SVM) algorithm achieved an accuracy of 76.9% when utilizing all 22 gait features, whereas the Naïve Bayes (NB) classifier reached a higher accuracy of 84.6% using four selected features¹¹. However, the study had a small sample size and relied on

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averaged data collected from noncontinuous gait analysis. This method does not fully replicate natural walking performance, which limits the reproducibility and generalizability of the findings.

As PD progresses, typical pathological symptoms, such as bradykinesia, rigidity, and impaired motor automaticity, become increasingly pronounced. This often necessitates higher dosages of antiparkinson medications^{14,15}. However, escalating dosages can lead to motor complications, including motor response fluctuations and dyskinesias¹⁵. The decision to initiate levodopa treatment is primarily influenced by the severity and rate of progression of the disease¹⁶. Accurate classification of the PD stage is crucial for optimizing therapeutic interventions, as different stages exhibit distinct motor and non-motor characteristics. Such classifications enable clinicians to tailor exercise interventions and medical treatments to meet the specific needs of patients at various stages of the disease¹⁶. Subtyping PD according to the disease stage could facilitate the development of personalized treatment protocols, potentially reducing dependence on medications associated with significant side effects, such as dyskinesias and motor fluctuations^{15,17}.

In a related study, Zhao et al. assessed gait performance using only the self-selected walking speed to detect the severity of PD (H&Y stages 2, 2.5, and 3)¹⁸. The results revealed that the ensemble K-nearest neighbor algorithm achieved an accuracy of 76.03%. However, the study had similar limitations, and because there is no consensus on using gait variables to detect PD severity, further research is necessary to explore gait variables that can help classify moderate-stages of PD. Lee et al. recommended assessing gait ability across various walking speeds commonly encountered in daily life to detect the severity of motor symptoms in individuals with PD¹⁹. These more challenging walking tasks may provide a more accurate assessment of gait variability, as they demand greater executive function and attention to maintain dynamic stability²⁰. Therefore, it is essential to consider challenging walking tasks under different speed conditions to identify markers of disease progression in individuals with PD. This study aimed to identify the accuracy of ML algorithms in classifying gait variables based on continuous gait tasks using three walking speeds, specifically to classify: i) people with PD versus age-matched healthy controls, ii) people with early PD (H&Y stages 1 to 2) versus age-matched healthy controls, and iii) people with moderate PD (H&Y stages 2.5 to 3) versus people with early PD (H&Y stages 1 to 2).

Methods

Participants

A total of 103 individuals with PD and 75 age-matched healthy controls participated in this study. The inclusion criteria for participants with PD were: (a) a clinician-confirmed diagnosis of idiopathic PD according to the UK Parkinson's Disease Society Brain Bank Clinical Diagnostic Criteria²¹, (b) age over 50 years, (c) a modified H&Y scale score of 1–3, indicating independent walking and mobility without the need for an assistive device, and (d) a score of ≥ 24 on the Korean Mini-Mental State Examination²². The control group consisted of age-matched healthy adults over 50 years old with no history of musculoskeletal, cardiovascular, vestibular, or neurological disorders that could have affected gait in the past six months (Fig. 1). The physical and clinical characteristics of all participants are presented in Table 1. Group 1 consisted of 75 age-matched healthy controls. Participants with PD were further categorized based on their H&Y scale: Group 2 included 61 those in stages 1 to 2, and Group 3 included 42 those in stages 2.5 to 3^{23,24}. This study was approved by the Institutional Review Board of Dong-A University Hospital (IRB number: DAUHIRB-22-089). All participants were informed of the experimental procedures and provided written informed consent, as approved by the committee. The study is registered in the Clinical Research Information Service of the Republic of Korea (KCT0009353).

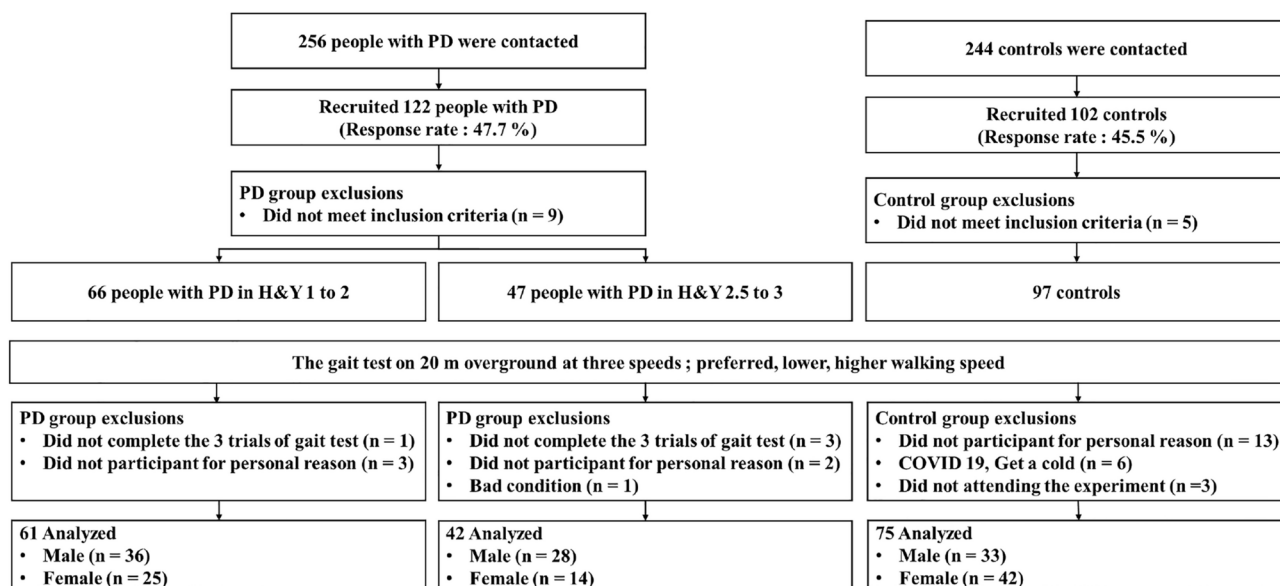


Fig. 1. Flow diagram for participants.

Characteristics	Controls (n = 75)	Early PDs (n = 61)	Moderate PDs (n = 42)	p-value
Sex (male/female)	33/42	36/25	28/14	0.050
Age (years)	68.91 ± 5.54	68.61 ± 6.24	69.45 ± 5.53	0.766
Height (cm)	160.61 ± 8.97	161.89 ± 8.46	160.15 ± 8.72	0.560
Body mass (kg)	63.94 ± 9.94	64.09 ± 9.28	61.65 ± 10.31	0.397
BMI (kg/m ²)	24.75 ± 3.13	24.41 ± 2.64	23.89 ± 2.29	0.279
K-MMSE (scores)	27.28 ± 1.80	27.75 ± 1.71	27.6 ± 1.53	0.253
Hoehn and Yahr scale (stages)	–	1.84 ± 0.37	2.69 ± 0.25	< 0.001^a
UPDRS total (scores)	–	52.00 ± 16.27	66.81 ± 17.83	< 0.001^b
UPDRS part III (scores)	–	32.84 ± 12.20	42.23 ± 13.07	< 0.001^b
PIGD (scores)	–	0.60 ± 0.34	1.17 ± 0.59	< 0.001^a
L-Dopa equivalent dose (mg/day)	–	571.89 ± 330.91	917.48 ± 519.66	< 0.001^a
Treatment duration (years)	–	4.93 ± 3.43	6.33 ± 3.30	0.020^a
Symptom duration (years)	–	5.36 ± 3.39	7.12 ± 3.76	0.012^a

Table 1. Physical and clinical characteristics of all participants. Significant values are in bold ($p < 0.05$). All data are indicated as mean ± standard deviation. PDs Parkinson’s disease, BMI Body mass index, K-MMSE Korean mini-mental state examination, UPDRS Unified Parkinson’s disease rating score, PIGD Postural instability/gait difficulty. ^aMann–Whitney U test. ^bIndependent sample T-test; $p < 0.05$.

Test procedures and data analysis

The gait analysis in this study was conducted using shoe-type data loggers (Smart Balance® SB-1, JEIOS, Korea) equipped with an inertial measurement unit (IMU) embedded in the shoe outsole²⁵. The IMU sensor captured triaxial acceleration (up to ± 6 g) and angular velocity (up to ± 500°/s) along three orthogonal axes. The measured data were transmitted via Bluetooth to a gait analysis system (DynaStab™, JEIOS, Korea). The spatiotemporal variables obtained from the shoe-type inertial measurement unit (IMU) sensor demonstrated excellent data agreement with a three-dimensional motion capture system, as evidenced by the intraclass correlation coefficients (ICCs) ranging from 0.944 to 0.985²⁶.

Participants with PD performed the gait test while in the “ON” state of antiparkinsonian medication for at least 2 h, reflecting their typical gait performance during daily activities. All participants were first assessed by a clinician for clinical characteristics using the Korean Mini-Mental State Examination, the H&Y scale, and the Unified PD Rating Scale. Physical characteristics, including height, weight, shoulder width, hip and waist circumference, thigh and calf circumference, and leg length, were also measured. Participants began with a 5-min warm-up involving stretching exercises before starting the experiment. Following the warm-up, the participants stood stationary on an overground surface while wearing a shoe-type IMU sensor to conduct static calibration. Subsequently, they performed a gait test along a 24-m straight walkway.

Each participant was instructed to walk at a comfortable pace, with a metronome used to measure their self-selected walking speed in steps per minute (beats/min). Lower walking speed (LWS) and higher walking speed (HWS) were defined as 80% and 120% of the preferred walking speed (PWS), respectively¹⁹. Prior to the test, participants completed three practice trials. After a 5-min break to wash out the effects of metronome-based pacing, participants performed the gait test at both HWS and LWS. All the walking speeds were measured using the metronome (beats per minute); however, the actual gait test was conducted without the metronome, as external auditory cues could affect the performance of individuals with PD²⁴ (Supplementary Fig. 1).

All the gait test data were collected at a frequency of 100 Hz and subsequently filtered using a second-order Butterworth low-pass filter with a cut-off frequency of 10 Hz. The spatiotemporal gait variables were analyzed over a 20-m section of the walkway, excluding the 2-m acceleration and deceleration phases at either end. During the gait cycle, gait events were defined based on heel strikes, which were identified when the linear acceleration reached its peak value along the anteroposterior axis. Moreover, toe-offs were determined to occur when linear acceleration peaked along the vertical axis^{26,27}. The spatiotemporal gait variables were calculated as follows: gait speed (m/s), stride length (m), single-support phase (%), double-support phase (%), stance phase (%), the coefficient of variation (CV) for all spatiotemporal variables²⁸, and gait asymmetry²⁹.

Statistical analysis

All the data were checked for normality using the Shapiro–Wilk test, and statistical analyses were performed using both parametric and nonparametric tests. The mean and standard deviation (SD) were calculated using one-way analysis of variance (ANOVA) for the physical characteristics of all participants, while independent t-tests and Mann–Whitney U tests were used to assess the clinical characteristics of individuals with PD. An intraclass correlation coefficient (ICC) analysis (2,1) was conducted to validate the agreement between the estimated and actual measured speeds under high- and low-speed conditions. The limits of agreement were calculated using a Bland–Altman plot to examine the error between the estimated and actual measured speeds³⁰.

To reduce the dimensionality of the input data, this study used the random forest (RF) method to assess the importance of the features. Specifically, the hyperparameters for the RF algorithms were tuned by experimenting with the number of trees, which were set to 100, 200, and 300, and the maximum number of features, which were set to 0.25, 0.5, and 0.75, while the random state was fixed at 42. This approach helps exclude the influence of

irrelevant variables and facilitates data post-processing. We evaluated the importance of variables in three cases: (i) between age-matched healthy controls and individuals with PD, (ii) between age-matched healthy controls and individuals with early-stage PD, and (iii) between individuals with early-stage PD and those with moderate-stage PD.

We ranked the variables in the order of importance for each case and selected the top 15 spatiotemporal gait variables. Finally, we used multivariate logistic forward stepwise regression analysis to identify the optimal gait variables to classify the three groups from these top 15 variables. The covariates included participant's age, gender, height, and body mass index. The cut-off values of the selected gait variables for group classification were examined using receiver operating characteristic (ROC) curves. The area under the ROC curve (AUC) was calculated to assess the ability of the selected variables to classify the groups³¹. The statistical significance was set at 0.05.

An ML approach was applied to explore the potential of spatiotemporal gait features to classify individuals with PD versus age-matched controls, as well as to distinguish between different PD severity levels. Several ML algorithms were employed to evaluate the suitability of these gait variables for classification across different groups, with each algorithm providing unique insights into the data, thus necessitating an assessment of the variation in the classification accuracy. To address the classification tasks, seven ML algorithms were applied (Fig. 2): logistic regression (LR)³², K-nearest neighbor (KNN)³³, NB³⁴, linear discriminant analysis (LDA)³⁵, quadratic discriminant analysis (QDA)³⁵, SVM³⁶, and random forest (RF)³⁷. The model hyperparameters for each classifier were tuned using a grid search (Table 2).

Five-fold cross-validation was used to evaluate the accuracy, precision, recall, and F1 scores for the classification performance of the seven algorithms. In this method, the dataset was divided into five parts, with four parts used as training data and one part used as test data in each iteration. The process was repeated five times, and the average of the results from the test sets was used as the final accuracy. We also calculated the Matthews correlation coefficient (MCC), which uniquely maximizes the sensitivity, specificity, precision, and negative predictive value, providing an accurate assessment of the overall performance of the algorithm on a given dataset. The MCC is often scaled to a normalized version (norm MCC) within the interval [0, 1], where 0 represents the worst performance and 1 represents the best performance³⁸. Given the imbalanced dataset, which included 75 individuals in Group 1, 61 in Group 2, and 42 in Group 3, a random oversampling method was employed to address this imbalance³⁹. Statistical analyses were performed using SPSS 21.0 (IBM Corp., Armonk, NY, USA) and Python (Python 3.10, Python Software Foundation).

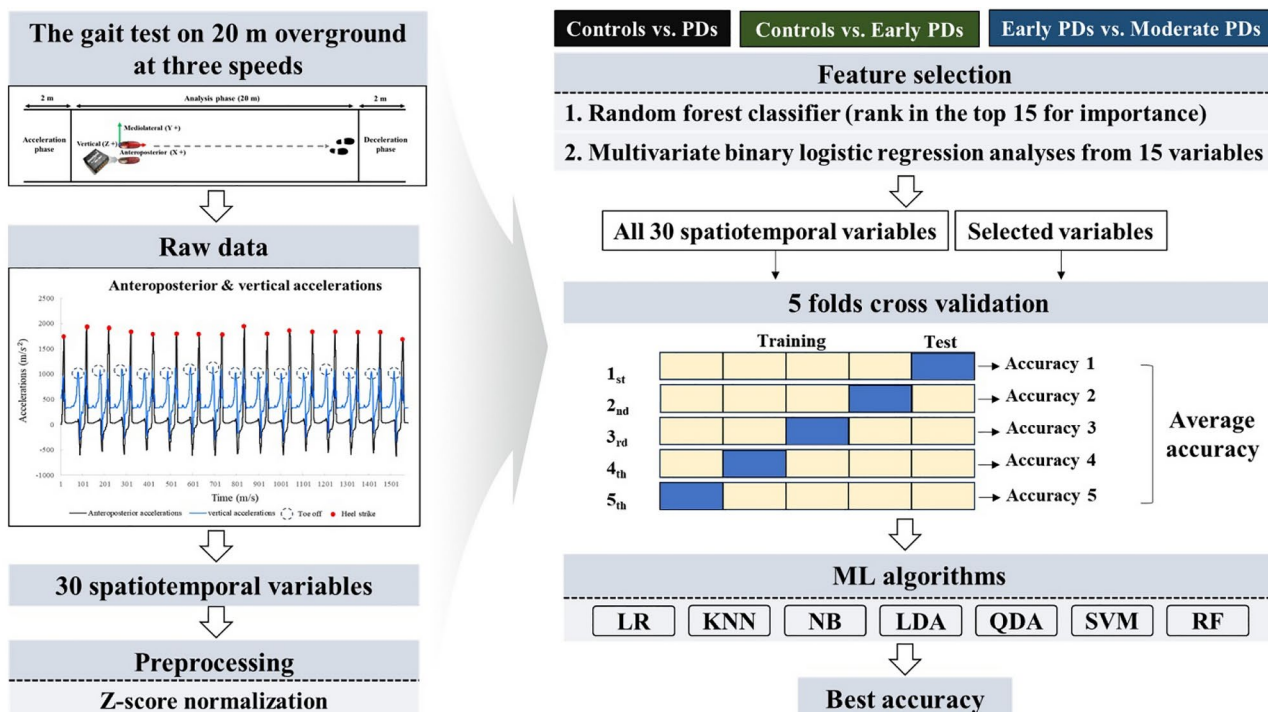


Fig. 2. Data process for machine learning analysis. *LR* Logistic regression, *KNN* Knearest neighbors, *NB* Naïve Bayes, *LDA* Linear discriminant analysis, *QDA* Quadratic discriminant analysis, *SVM* Support vector machine, *RF* Random forest, *PDs* people with PD, *Early PDs* people with PD in the early-stage, *Moderate PDs* people with PD in the moderate-stage.

MLs	Controls vs. PDs (30 Features)	Controls vs. PDs (3 Features)	Controls vs. Early PDs (30 Features)	Controls vs. Early PDs (2 Features)	Early PDs vs. Moderate PDs (30 Features)	Early PDs vs. Moderate PDs (1 Feature)
LR	$C=1.0$	$C=1.0$	$C=0.001$	$C=0.1$	$C=1.0$	$C=0.1$
KNN	$k=9$	$k=9$	$k=8$	$k=7$	$k=3$	$k=7$
NB	–	–	–	–	–	–
LDA	$n_components=1$	$n_components=1$	$n_components=1$	$n_components=1$	$n_components=1$	$n_components=1$
QDA	$reg_param=0.3$	$reg_param=0.001$	$reg_param=0.001$	$reg_param=0.4$	$reg_param=0.1$	$reg_param=0.5$
SVM	$C=7.6$, $gamma=scale$, $kernel=linear$	$C=29.5$, $gamma=0.01$, $kernel=rbf$	$C=1.9$, $gamma=0.001$, $kernel=rbf$	$C=1719$, $gamma=0.0001$, $kernel=rbf$	$C=0.5$, $gamma=1.0$, $kernel=rbf$	$C=7.6$, $gamma=0.01$, $kernel=rbf$
RF	$max_depth=10$, $n_estimators=1250$	$max_depth=30$, $n_estimators=1250$	$max_depth=15$, $n_estimators=1500$	$max_depth=15$, $n_estimators=500$	$max_depth=30$, $n_estimators=750$	$max_depth=10$, $n_estimators=500$

Table 2. Model parameters of seven classifiers estimated by grid search. *ML* Machine learning, *PDs*, People with Parkinson's disease, *LR* Logistic regression, “*C*” is the inverse of the regularization strength; *KNN* K-nearest neighbor, “*k*” is the number of neighbors; *NB* Naïve Bayes, *LDA* Linear discriminant analysis, “*n_components*” is the number of components, *QDA* Quadratic discriminant analysis, “*reg_param*” is the regularization of the per-class covariance, *SVM* Support vector machine, “*C*” is the regularization parameter, and “*gamma*” is the kernel coefficient; *RF* Random forest, “*n_estimators*” is the number of trees in the forest, and “*max_depth*” is the maximum depth of the tree.

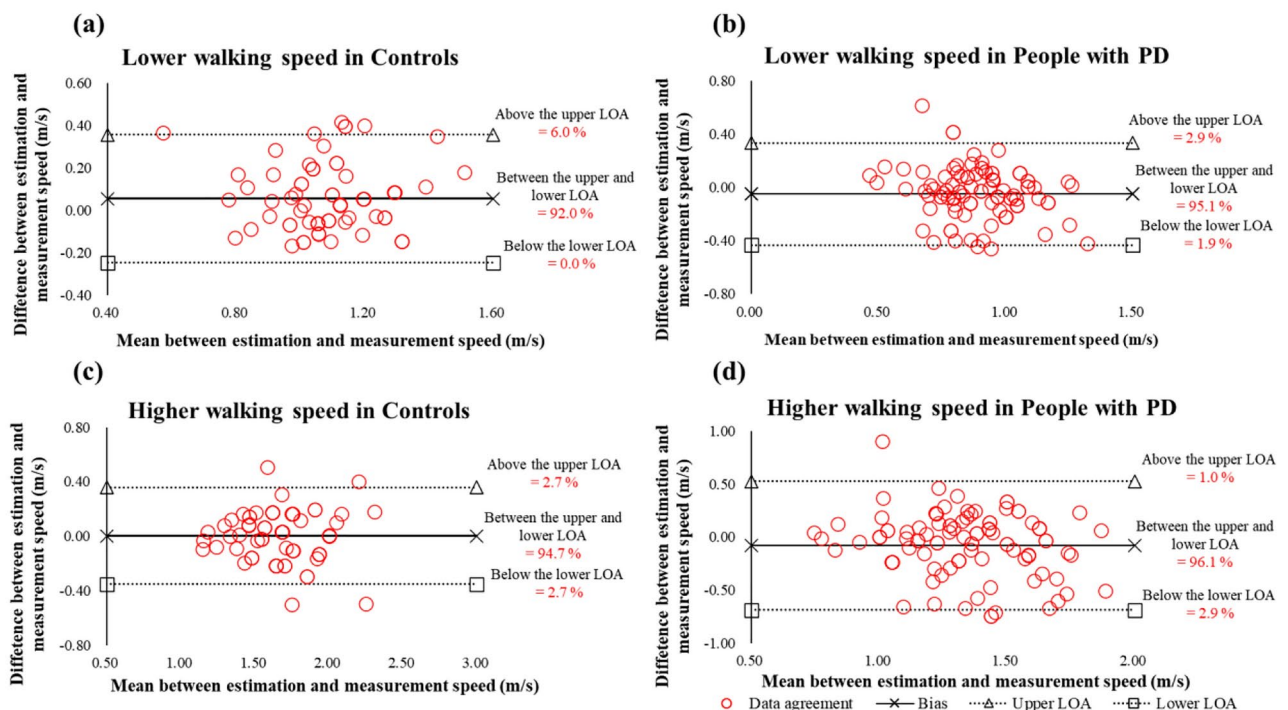


Fig. 3. Bland–Altman plots for data agreement between estimated and measured walking speeds. (a) and (b) lower speed in controls and people with PD, (c) and (d) higher speed in controls and people with PD; LOA: limits of agreement.

Results

Reliability of estimated and measured speeds

Supplementary Table S1 presents the reliability of the estimated and measured walking speeds at the LWS and HWS for both the control group and individuals with PD. The agreement between the estimated and measured walking speeds ranged from 92.0% to 95.1% at LWS and from 94.7% to 96.1% at HWS for both groups (Fig. 3).

Feature selection for group classification

The feature importance values for all 30 gait variables in each of the three classification cases are listed in Supplementary Table S2 (see Supplementary Tables S3–S5 for a comparison of the gait variables across classification cases), including the ranking of the importance of all the variables, highlighting those ranked in the top 15 for each case (Fig. 4, Supplementary Fig. 2). Using these selected variables, the multivariate logistic

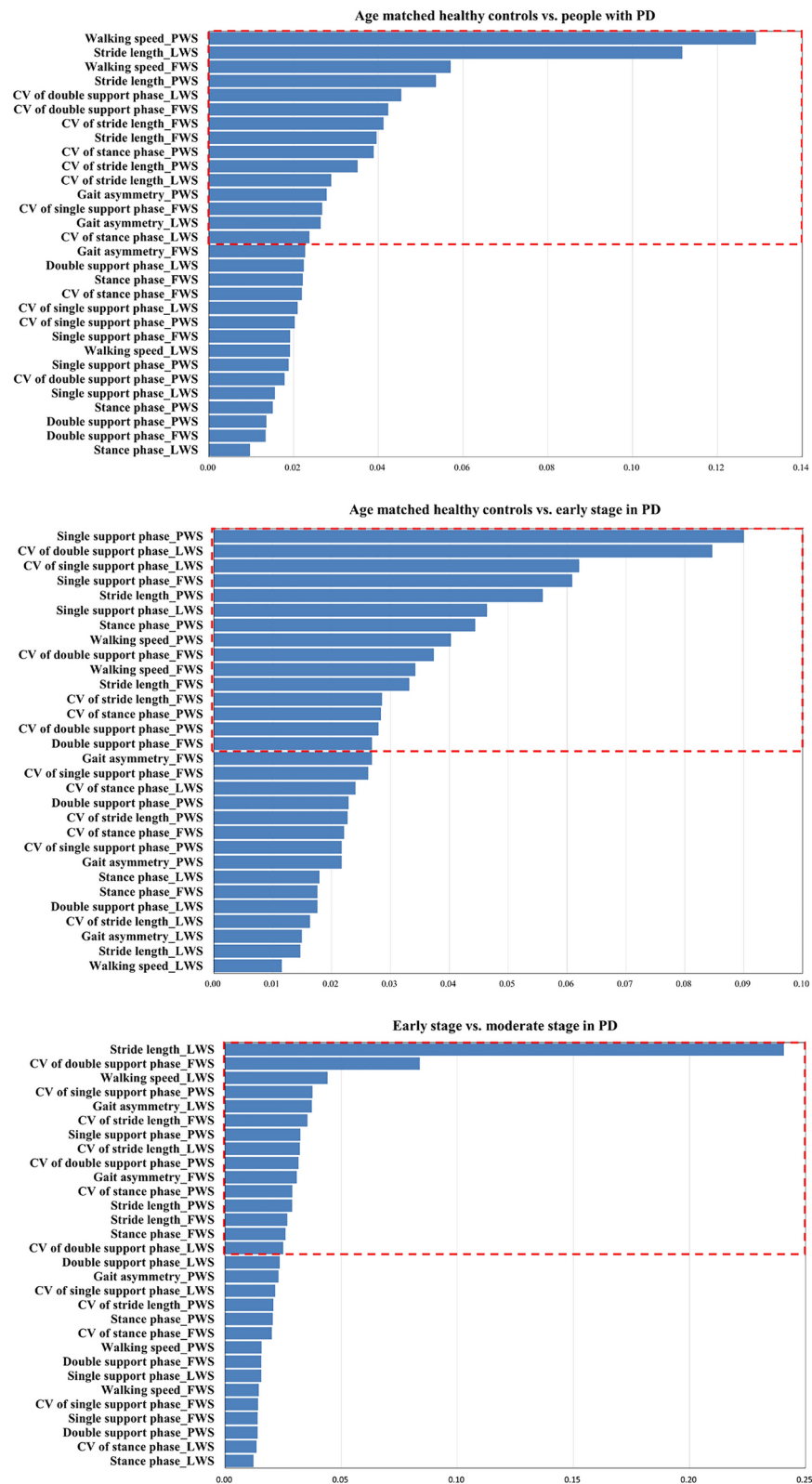


Fig. 4. Results of all 30 variables importance. The red dotted line surrounds the variables that were ranked in the top 15; LWS Lower walking speed, PWS Preferred walking speed, HWS Higher walking speed, CV Coefficient of variance.

Variables	Estimate	SE	OR	95% CI for OR	p-value	R _N ²
Controls and PDs						
CV of stride length at the LWS (%)	0.599	0.256	1.820	1.101–3.007	0.020	0.528
Walking speed at the PWS (m/s)	−0.864	0.392	0.421	0.195–0.909	0.027	
Stride length at the HWS (m)	−1.642	0.460	0.194	0.079–0.477	<0.001	
Controls and early PDs						
CV of stride length at the LWS (%)	0.535	0.240	1.707	1.067–2.732	0.026	0.404
Stride length at the HWS (m)	−1.790	0.364	0.167	0.082–0.341	<0.001	
Early PDs and moderate PDs						
Walking speed at the PWS (m/s)	−0.979	0.282	0.376	0.216–0.653	0.001	0.291

Table 3. Results of binary logistic regression for different groups analyzed. Dependent variable 1 = Controls, 2 = PDs; 1 = Controls, 2 = H&Y 1–2; 1 = H&Y 1–2, 2 = H&Y 2.5–3. This model adjusted for age, gender, height, BMI, and MMSE score; PDs: People with Parkinson’s disease; Early PDs: H&Y 1–2; Moderate PDs: H&Y 2.5–3; CV Coefficient of variance, PWS Preferred walking speed, LWS Lower walking speed, HWS Higher walking speed, SE Standard error, OR Odds ratio, CI Confidence interval; R_N² is the fit statistic for the Nagelkerke model; *p* < 0.05.

MLs	Controls vs. PDs (30 Features)	Controls vs. PDs (3 Features)	Controls vs. Early PDs (30 Features)	Controls vs. Early PDs (2 Features)	Early PDs vs. Moderate PDs (30 Features)	Early PDs vs. Moderate PDs (1 Feature)
LR	67.9 ± 9.1	69.4 ± 6.7	65.3 ± 4.5	66.0 ± 4.3	64.8 ± 14.1	68.1 ± 9.3
KNN	64.6 ± 6.4	67.9 ± 6.4	62.7 ± 5.5	64.0 ± 6.0	70.5 ± 11.6	63.9 ± 11.9
NB	66.5 ± 7.8	68.9 ± 7.7	63.3 ± 11.3	67.3 ± 8.6	56.7 ± 12.9	69.0 ± 10.6
LDA	65.5 ± 8.9	68.4 ± 8.7	64.7 ± 9.3	64.7 ± 3.0	57.5 ± 8.7	68.1 ± 9.3
QDA	68.9 ± 6.5	69.9 ± 7.9	66.7 ± 5.3	66.0 ± 8.3	64.8 ± 6.9	69.8 ± 11.4
SVM	66.0 ± 8.1	66.5 ± 9.0	67.3 ± 4.9	64.7 ± 3.8	75.5 ± 5.4	69.8 ± 11.4
RF	79.1 ± 5.4	78.1 ± 9.0	71.3 ± 6.9	61.3 ± 11.2	75.4 ± 6.4	66.5 ± 10.2

Table 4. Accuracies of seven classifiers from five-fold cross-validation. Mean (%) ± standard deviation (%) is calculated from the five-fold cross-validation; the mean values presented in boldface denote the best performance (the highest test accuracy); MLs Machine learning techniques, LR Logistic regression, KNN K-nearest neighbors, NB Naïve Bayes, LDA Linear discriminant analysis, QDA Quadratic discriminant analysis, SVM Support vector machine, RF Random forest.

forward stepwise regression analyses revealed significant differences between the control group and individuals with PD. Specifically, the CV for stride length at LWS (cut-off value: 2.64%; AUC: 0.409; sensitivity: 0.453; specificity: 0.456; odds ratio [OR]: 1.820), walking speed at PWS (cut-off value: 1.27 m/s; AUC: 0.781; sensitivity: 0.747; specificity: 0.738; OR: 0.421), and stride length at HWS (cut-off value: 1.43 m; AUC: 0.757; sensitivity: 0.680; specificity: 0.680; OR: 0.460) showed significant differences. In the classification between controls and individuals with early-stage PD, significant differences were observed in the CV of stride length at LWS (cut-off value: 2.62%; AUC: 0.405; sensitivity: 0.453; specificity: 0.459; OR: 1.707) and stride length at HWS (cut-off value: 1.48 m; AUC: 0.690; sensitivity: 0.613; specificity: 0.656; OR: 0.167). Additionally, in the classification between individuals with early-stage PD and those with moderate-stage PD, walking speed at PWS (cut-off value: 1.18 m/s; AUC: 0.739; sensitivity: 0.590; specificity: 0.738; OR: 0.376) showed significant differences (Table 3).

ML approach

This study addressed three classification tasks using two distinct feature sets: one comprising all spatiotemporal features and the other comprising features selected through binary stepwise regression (Table 4, Fig. 5). In the classification of controls versus individuals with PD, the RF algorithm demonstrated the highest accuracy when utilizing all the features (79.1% ± 5.4% SD; norm MCC: 0.79). Following feature reduction through stepwise regression, RF continued to exhibit the highest accuracy (78.1% ± 9.0% SD; norm MCC: 0.78). For the early-stage classification of controls versus individuals with PD, the RF again achieved the highest accuracy with all the features (71.3% ± 6.9% SD; norm MCC: 0.72), whereas the NB algorithm yielded the highest accuracy (67.3% ± 8.6% SD; norm MCC: 0.68) using two selected features. In distinguishing between early-stage and moderate-stage PD, the SVM achieved the highest accuracy when using all the features (75.5% ± 5.4% SD; norm MCC: 0.79), while both the QDA and SVM attained the highest accuracy (69.8% ± 11.4% SD; norm MCC: 0.70) with a single selected feature. Moreover, classification performance metrics, such as the recall, precision, and F1 score, were evaluated using a confusion matrix (Supplementary Table S6, Fig. 6).

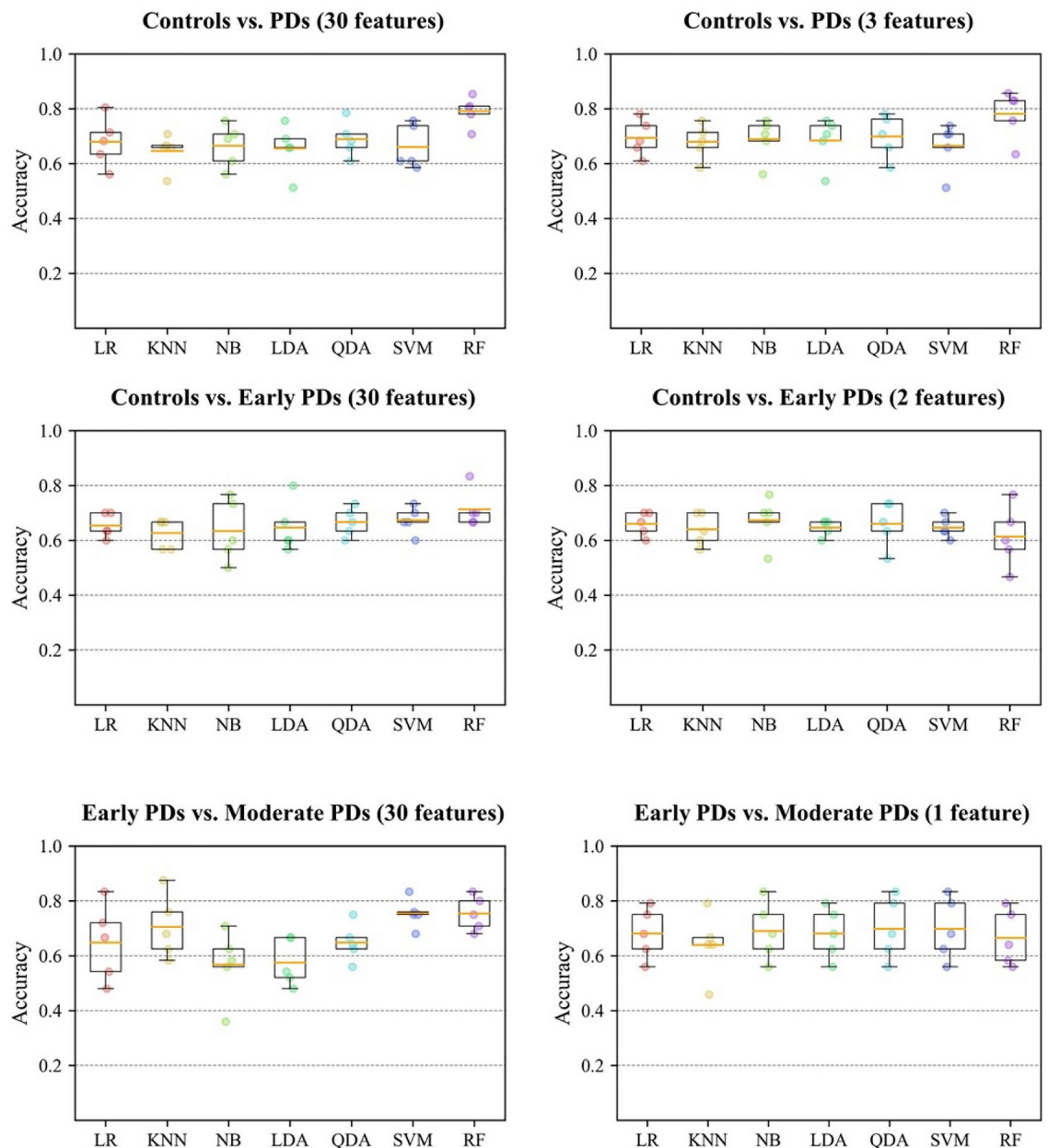


Fig. 5. Accuracies of seven classifiers. Orange line in box plot indicates mean values. *LR* Logistic regression, *KNN* Knearest neighbors, *NB* Naïve Bayes, *LDA* Linear discriminant analysis, *QDA* Quadratic discriminant analysis, *SVM* Support vector machine; *RF*: Random forest.

Discussion

Classification of people with PD and age-matched healthy controls

This study classified age-matched healthy controls and individuals with PD using gait variables under LWS, PWS, and HWS conditions. We demonstrated that individuals with PD exhibited a 57.9% reduction in walking speed at PWS and a 54.0% shortening of stride length at HWS compared to controls, which is consistent with previous reports^{40,41}. Recently, Trabassi et al. evaluated the classification performance of ML models applied to individuals with PD and healthy controls, with the SVM achieving an accuracy of 81% for all 22 features and 86% for seven selected features, including the CV of the step length, stride length, and stance phase⁴². In comparison, the present study achieved an accuracy of 79.1% for all 30 features and 78.1% for the three selected features (CV of the stride length at LWS, walking speed at PWS, and stride length at HWS) using the RF algorithm, demonstrating similar performance to previous findings.

Individuals with PD exhibit disrupted walking speeds, primarily due to difficulties in generating adequately sized steps and delays in movement timing, which are attributed to impaired cue production in the basal ganglia. This disruption may also involve the under-scaling of movement through the cortical-thalamic-basal ganglia circuitry⁴³. However, none of the previous studies considered variable walking speeds or highlighted

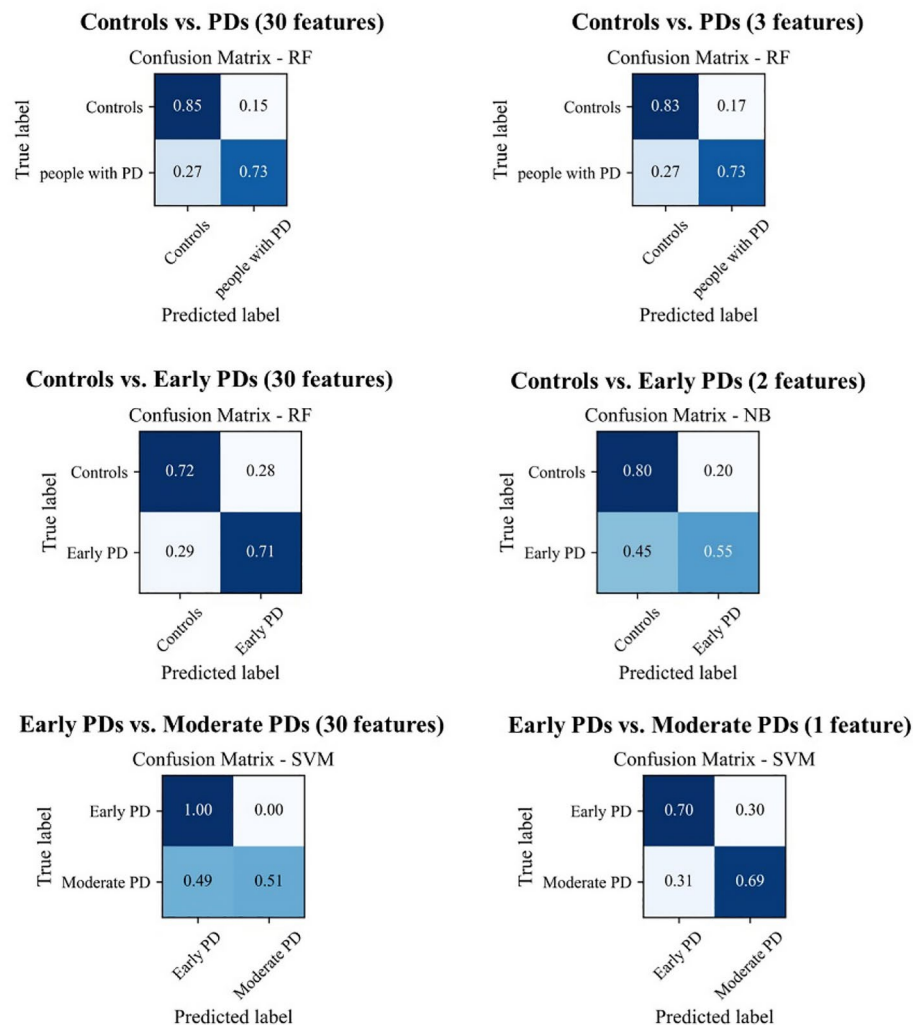


Fig. 6. Confusion matrices for six cases. *PDs* people with PD, *RF* Random forest, *NB* Naïve Bayes, *SVM* Support vector machine.

the importance of gait variability in their analyses. The result could be explained by the fact that walking at a preferred speed represents a gait pattern that minimizes variability and requires the least energy expenditure, regardless of cerebellar motor dysfunction or vestibular feedback control disorders⁴³.

In this study, we identified the variability of stride length at HWS as an important variable for distinguishing individuals with PD from controls. Increased variability in individuals with PD is thought to stem from a reduced ability to control gait timing, resulting from dysfunction in various components of the basal ganglia and other regions of the locomotor network. This dysfunction leads to a series of disjointed strides, which ultimately decrease walking speed². Furthermore, both LWS and HWS are associated with prolonged single-limb stance phases and lateral displacement of the center of mass, further contributing to increased gait variability in individuals with PD^{44,45}. Collectively, individuals with PD may experience difficulty walking under both slow and fast conditions, which can sensitively detect a decline in walking ability. Therefore, we suggest that evaluating gait under slower, faster, and preferred walking speeds may provide a valuable approach for overcoming these limitations⁴⁴.

Classification of controls and people with early-stage PD

As a primary finding, the classification between individuals with early-stage PD and age-matched healthy controls, using multivariate logistic forward stepwise regression analysis, identified two significant gait features: the CV of the stride length at LWS (OR: 1.707; AUC: 0.405) and shortened stride length at HWS (OR: 0.167; AUC: 0.690), with an explanatory power of 40.4%. Additionally, the RF algorithm achieved an accuracy of 71.3% using all features, while NB reached an accuracy of 67.3% with the two selected features.

In this study, individuals with early-stage PD demonstrated a 1.71-fold increase in the CV of stride length at LWS and an 83.3% reduction in stride length at HWS compared to age-matched healthy controls, consistent with previous studies^{9,10,46}. Motor symptoms in early-stage PD often manifest as unilateral parkinsonism, primarily affecting one side of the body, due to asymmetrical degeneration of the basal ganglia¹⁴. Impaired basal ganglia

function can result in reduced stride length during all gait phases due to bradykinesia and decreased walking velocity in individuals with early-stage PD¹⁴.

A recent study by Ferreira et al. performed gait analysis on an 8.5 m walkway at self-selected walking speeds to classify individuals with early-stage PD (H&Y stages 1 and 2)¹¹. The study found that SVM with a radial basis function kernel achieved an accuracy of 76.9% using all 22 features, while NB achieved an accuracy of 84.6% using four selected features, including walking speed, step length, step width, and the CV of step width. They suggested that the pace and variability domains of walking are the most discriminative features in classifying early-stage PD, which aligns with the findings of this study. In our study, the CV of step length at HWS and step length at LWS achieved a lower accuracy (67.3% using the NB algorithm) compared to the previous study, which used non-continuous gait analysis data. Non-continuous steps may better mimic natural walking performance but can limit reproducibility and generalizability⁴⁷. Some researchers suggest that gait assessments on a 20 m walkway, with at least 30 steps, can improve the reliability and generalizability of the results^{44,48}. Therefore, conducting gait assessments with at least 30 consecutive steps under various speed conditions may provide a more objective and sensitive method for classifying early-stage PD from age-matched healthy controls.

Classification of early and moderate-stages of PD

In the classification between individuals with early-stage PD and those with moderate-stage PD, the selected gait feature was walking speed at PWS (OR: 0.376; AUC: 0.739), with an explanatory power of 29.1%. Additionally, the SVM algorithm achieved an accuracy of 75.5% using all features, while both SVM and QDA achieved an accuracy of 69.8% using the single selected feature.

As PD progresses, motor symptoms, such as bradykinesia, rigidity, and bilateral involvement, become more pronounced, reducing gait asymmetry compared with early-stage PD¹⁴. In the present study, the walking speed at PWS was 1.707 times lower in individuals with moderate-stage PD compared with those with early-stage PD. This finding highlights the significant discriminative power of walking speed in classifying individuals with PD between early and moderate-stages. Although Keloth et al. (2020) reported no significant differences in the mean values of the gait parameters, including the stride interval, swing interval, stance interval, and double-support interval⁴⁶, our results indicate that the walking speed deteriorates significantly as PD progresses. This finding is consistent with previous results⁴⁹. This emphasizes the importance of slower movement as a key independent factor in walking ability during the early-to moderate-stages of PD⁵⁰.

Recent classification studies using ML models have not specifically investigated the accuracy of classifying individuals with PD across H&Y stages 1 to 3 based on spatiotemporal gait features from straight walkways. In this study, we achieved an accuracy of 69.8% for walking speed at PWS, a key feature in moderate-stage PD. Although this accuracy is slightly lower than that reported in previous studies^{10,51}, the use of sufficient consecutive steps allowed us to achieve reliable accuracy in classifying PD in the early-to moderate- stages⁴⁴. Thus, gait assessments involving at least 30 consecutive steps under various speed conditions may provide a more objective and sensitive method for classifying individuals with PD across these stages.

Clinical implications

The study results provide valuable clinical significance. First, a digital marker based on three walking speeds, requiring at least 30 consecutive steps, has been identified for H&Y stage classification. This digital marker may assist in distinguishing early-stage PD from normal aging and in classifying the severity of motor symptoms in PD. Furthermore, it has the potential to significantly impact clinical practice by facilitating personalized therapeutic and exercise interventions tailored to individual patient characteristics. The study identified key features for distinguishing individuals with early-stage PD, including stride length and the CV of stride length, which are associated with reduced strength, increased rigidity, and instability. These individuals in the early stages of PD may particularly benefit from interventions aimed at improving lower limb movement and timing, such as treadmill training or rhythmic auditory stimulation^{52,53}. Walking speed is widely recognized as another key indicator of functional activity in daily life⁵⁴. Our findings indicate that walking speeds are slower for individuals with moderate-stage PD than for those in the early stages. This decline in walking speed may reflect diminished muscle strength and a reduced capacity for rapid muscle contractions⁵⁵. These observations underscore the potential of targeted interventions, such as progressive resistance training, to address walking speed, mitigate bradykinesia, and lower the risk of falls in moderate-stage PD⁵⁶. Moreover, utilizing selected gait features across three walking speeds may help clinicians save time in assessing motor symptoms while providing a more objective evaluation of the gait ability in clinical settings.

Study limitations

This study has several limitations. First, given that most individuals with PD typically perform daily activities in the ON state⁵⁷, we analyzed the gait ability solely during dopaminergic medication intake to reflect the practical scenarios faced by individuals with PD^{58,59}. However, for a more comprehensive understanding of the motor performance, future studies should assess the gait performance in both the ON and OFF medication states. Second, our cohort exhibited an imbalance in both gender and the number of participants across groups, which may have influenced the statistical significance of our findings. In our analysis, we adjusted for this imbalance by including gender as a covariate in a multivariate logistic forward stepwise regression analysis. Nevertheless, similar analyses should consider the number of participants in each group and prioritize the recruitment of age- and gender-matched cohorts. Moreover, we examined a limited dataset of individuals with early and moderate-stage PD, which may have impacted the performance of the ML models⁶⁰. A larger sample size could enhance the reliability of ML algorithms and facilitate the use of advanced ML techniques and deep learning, which require substantial datasets for effective training⁶¹. Recent methodologies, including the Mahalanobis Metric-Based Oversampling Technique and the Conditional Tabular Generative Adversarial Network, have shown promise in

addressing challenges associated with small and imbalanced datasets. As demonstrated in previous studies^{62–64}, these techniques improve data representation and mitigate biases, thereby enhancing ML model performance. Consequently, incorporating such approaches in future research could provide valuable insights for improving model outcomes in unbalanced datasets. Third, while the ML algorithms demonstrated promising results in classifying age-matched healthy controls and individuals with PD, their performance in distinguishing between early- and moderate-stage PD was somewhat lower. This reduced accuracy likely stems from the challenges ML algorithms face when trained on small datasets, which often lack the diversity needed to generalize patterns effectively⁶⁵. Consequently, developing strategies to overcome the limitations of small datasets is both scientifically and practically significant. Transfer learning represents a particularly promising approach for improving generalization in cases with a limited number of training samples⁶⁶, as it leverages models pre-trained on large-scale source domains, such as image or time-series data. However, applying transfer learning in this study was challenging due to the nature of the extracted features used⁶⁷. Future research should explore transfer learning techniques to enhance generalization and robustness. In addition, incorporate multimodal datasets that encompass clinical, cognitive, motor, and neuroimaging variables, which may help improve the reliability and accuracy of characterizing PD progression^{60,68}. Lastly, while we utilized the H&Y scale as the primary method for classifying PD, future research should also consider both the H&Y scale and the UPDRS Part III, which can capture more intricate details of the disease⁶⁹.

Conclusions

This study analyzed the accuracy of ML algorithms in classifying gait variables based on a continuous gait task performed at three different walking speeds, aimed at early detection and severity classification of PD. The key gait markers for distinguishing individuals with PD from healthy controls were walking speed at PWS, stride length at HWS, and the CV of stride length at LWS. These features achieved a classification accuracy of 78.1% using the RF algorithm. For early detection of PD, stride length at HWS and the CV of stride length at LWS demonstrated strong potential, achieving an accuracy of 67.3% with the NB algorithm. Furthermore, walking speed at PWS was the most effective feature for assessing disease severity, achieving an accuracy of 69.8% using SVM and QDA.

These findings suggest that assessing gait over at least 30 consecutive steps at three different walking speeds could provide an objective and reliable method for evaluating motor symptoms in PD. Such assessments could serve as a valuable tool for clinicians and rehabilitation professionals, offering insights that support the development of personalized treatment and rehabilitation strategies tailored to the progression of the disease.

Data availability

The datasets supporting this study's findings are available from the corresponding author upon reasonable request.

Code availability

There is no available open-source code. The code for training and testing the ML models was written in Python 3.8 using PyTorch 1.9.1 and Torchvision 0.10.1. Data management and feature processing scripts were written in Python 3.8 using Pandas 1.3.3 and NumPy 1.21.2. The datasets supporting this study's findings, as well as the code used for the analysis, are available from the corresponding author upon reasonable request.

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Author contributions

JH, CY, HP, BK, HC, and SC conceived and designed the study. JH, HP, BK, HC, and SC recruited the participants. JH, CY, HP, BK, HC, and SC performed the data acquisition. JH, CY, HP, BK, HC, and SC analyzed and interpreted the data. JH, CY, HP, BK, HC, and SC drafted the article. All authors read and approved the final version of the manuscript submitted.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics declarations

All study procedures involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. The study protocol and supplementary information files were approved by the Institutional Review Board of Dong-A University Hospital (IRB number: DAUHIRB-22-089) (see ethics approval letter in the supplementary file). All patients provided written informed consent before data collection. The study is registered with the Clinical Research Information Service in the Republic of Korea (KCT0009353).

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-83975-3>.

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